Exercises in Marine Ecological Genetics

02. Hardy-Weinberg equilibrium

- Calculate allele and genotype frequencies
- Test for Hardy-Weinberg equilibrium
- Work with genotype data

Martin Helmkampf

https://github.com/mhelmkampf/meg26

Course outline

Preliminary, may be subject to change

Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	Population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II (optional)	11_edna.R

Today's objectives

By the end of this session, you will be able to:

- **Perform basic population genetic analyses in R**, including importing and handling genotype data (e.g., Genepop format)
- Calculate allele and genotype frequencies, and test for **Hardy-Weinberg equilibrium**
- Interpret how **locus-specific and genome-wide deviations** from HWE can reflect biological processes such as selection, population structure, or inbreeding

Accessing the course materials

All code and data for the course will be provided through a **Git repository** (please bookmark):

https://github.com/mhelmkampf/meg26

```
meg26
├── code
├── data
├── pdf
└── work
```

Connecting to the cluster via SSH

```
### Establish SSH connection to cluster using your course account
ssh user1234@rosa.hpc.uni-oldenburg.de  # please adjust user name

### Enter password (invisible)
```

...

```
### Either: Download course materials to course account using Git (first time)
git clone https://github.com/mhelmkampf/meg26.git

### Or: Update git repository (if it already exists)
cd meg25
git pull
```

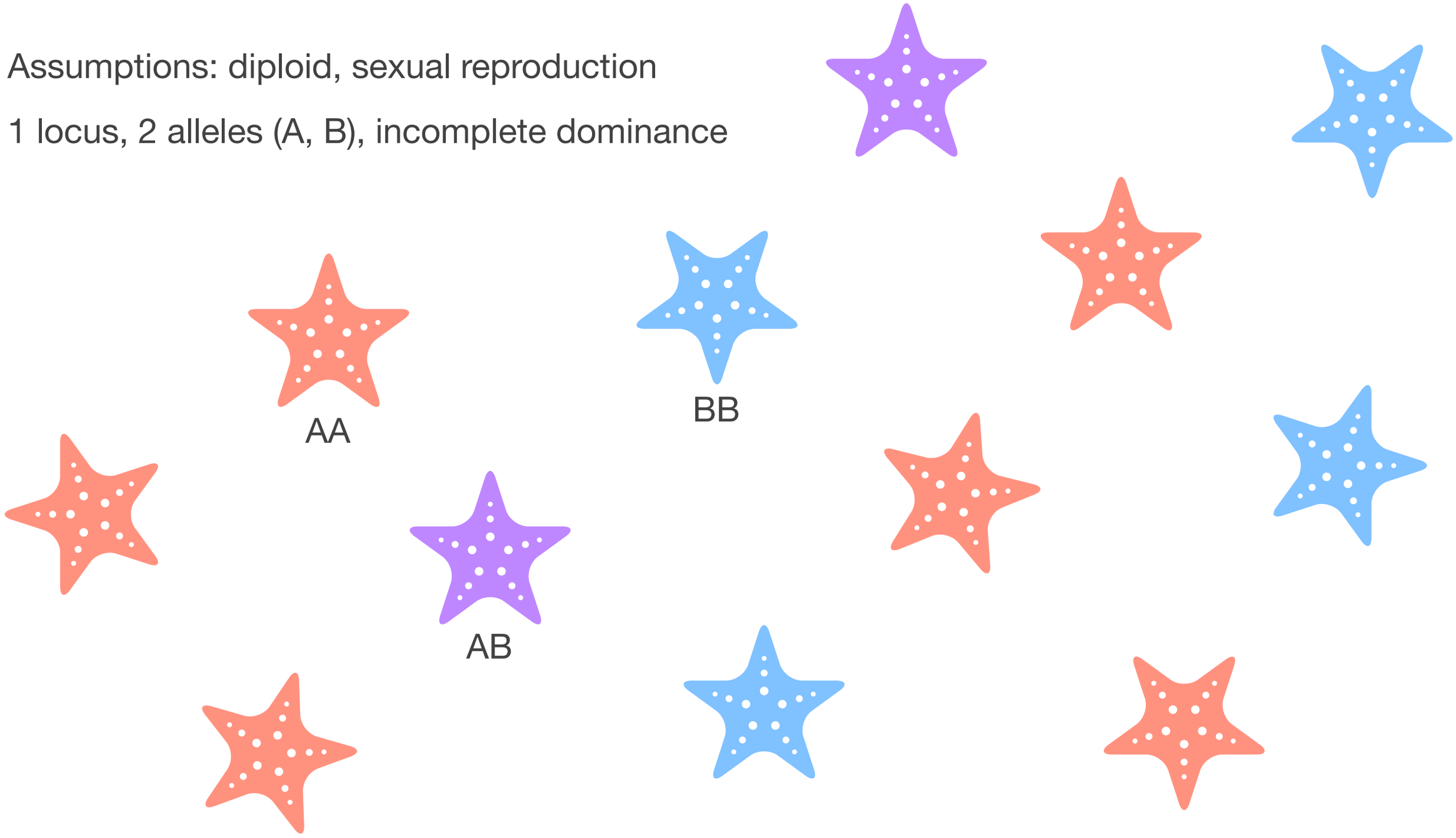
Keep following instructions in course script (02_hwe.R)

Is this population in Hardy-Weinberg equilibrium?

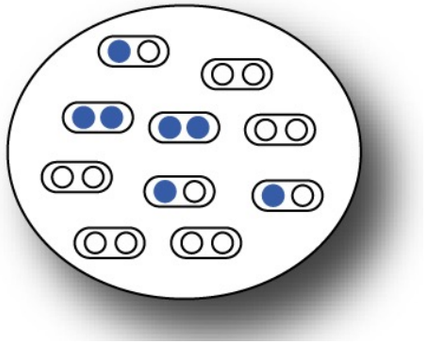
Exercise 1

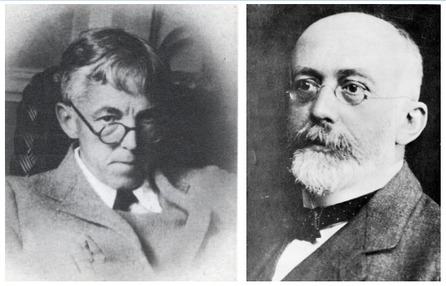
Assumptions: diploid, sexual reproduction

1 locus, 2 alleles (A, B), incomplete dominance



HARDY-WEINBERG (1908)





Godfrey H. Hardy (1877-1947)

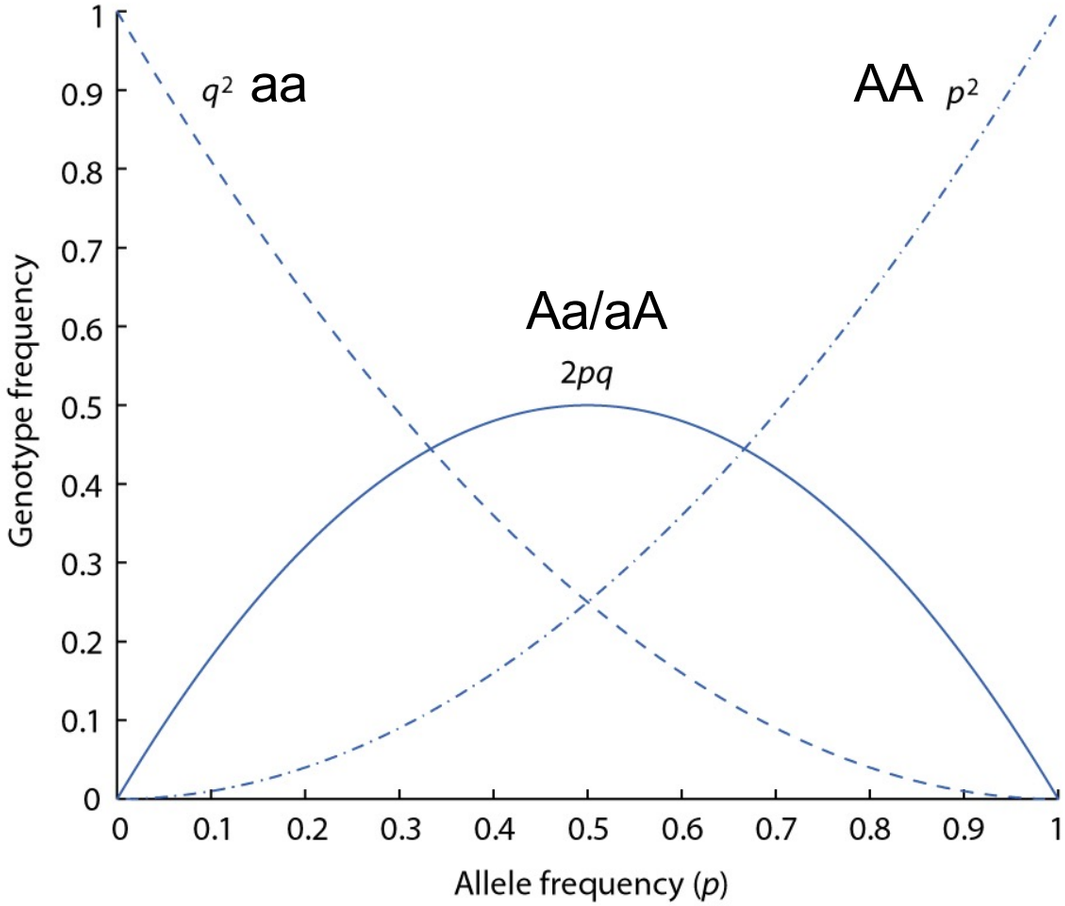
Wilhelm Weinberg (1862-1937)

Establish the relationship between allele frequencies and genotype frequencies in a population

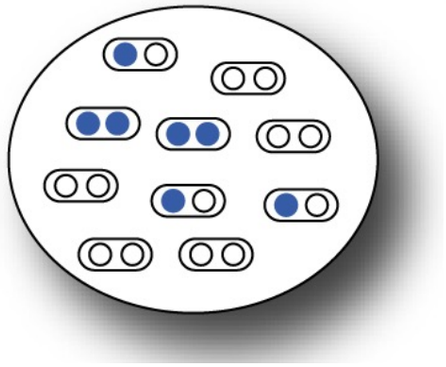
$$p^2 + 2pq + q^2 = 1$$

AA Aa/aA aa

p and *q*: allele frequencies for a locus with two alleles (A and a) ($p + q = 1$)



HETEROZYGOSITY



In one population

H_o = proportion of heterozygote individuals, observed heterozygosity

$H_e = 2pq = 1 - p^2 - q^2$, expected heterozygosity (assuming HW equilibrium)

$$F = \frac{H_e - H_o}{H_e}$$

Fixation index: *proportion by which heterozygosity is reduced or increased relative to the heterozygosity of a population at HW equilibrium with the same allele frequencies*.

Divided by H_e −> *proportion* (of expected heterozygosity)

Varies between -1 and 1

$F < 0$: heterozygote excess

$F > 0$ heterozygote deficit (homozygote excess)

May be averaged over several loci −> reduces bias

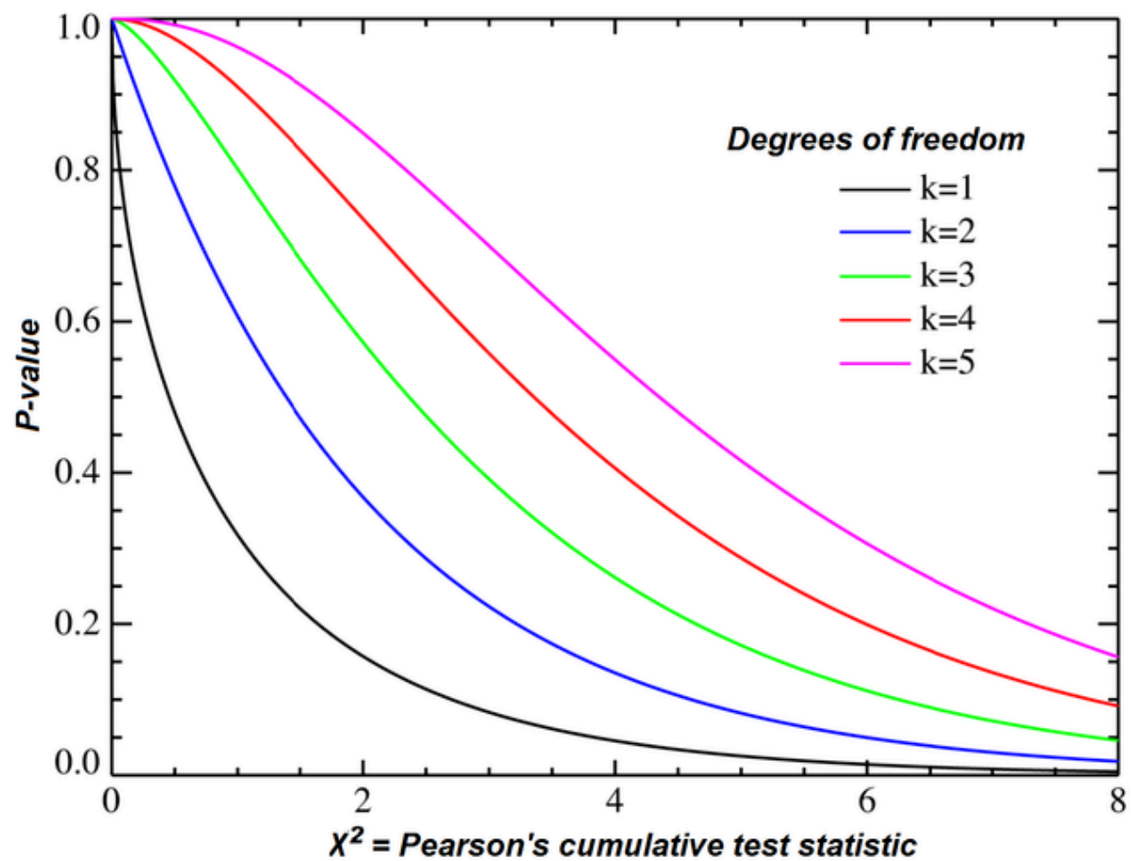
May be extended to *k* alleles

Pearson's chi-squared test

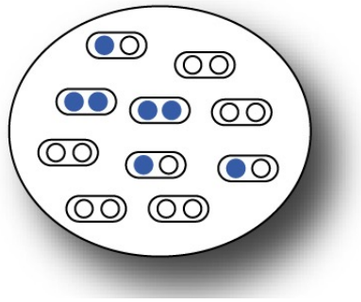
Chi-square statistic:

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Chi-square distribution:



A *single generation of reproduction will result in a population that meets the expected Hardy-Weinberg frequencies*, i.e. is at *Hardy-Weinberg* (HW) *equilibrium*



Assuming an “ideal” population, i.e. :

- Diploid organisms
- Sexual reproduction (as opposed to clonal)
- Random mating (as opposed to e.g. assortative) with respect to genotype
- Random union of gametes
- Discrete, non-overlapping generations
- Very large (infinite) population
- No migration
- No population structure
- No natural selection
- Two alleles
- Identical allele frequencies in both sexes

-> Departures from HW equilibrium may indicate:

- Inbreeding
- Assortative mating
- Self-fertilization
- Natural selection
- Population structure
- ...

Genepop file format

puella_barbados.txt

```
Microsatellite genotypes of Hypoplectrus puella from Barbados
g2
gag010
h24
hyp001
hyp015
hyp018
e2
hyp008
hyp016
pam013
POP
barbados ind#735 , 203235 121127 210248 225231 126132 190192 157157 236244 194220 114128
barbados ind#736 , 205217 119119 204216 225229 126126 190192 157165 236236 216222 106144
barbados ind#737 , 203203 135147 224226 233235 126136 190198 157163 238250 202226 126160
barbados ind#738 , 211217 119121 228230 223231 126130 190190 157157 240244 220226 138160
barbados ind#739 , 205225 121125 208216 227231 132132 192192 158159 244256 192196 146146
barbados ind#740 , 217233 119121 216228 227233 126126 192198 157157 242256 194216 000000
barbados ind#741 , 203209 000000 216222 223229 126130 192198 157157 234236 188226 118130
barbados ind#742 , 203215 119119 222234 233233 130130 190190 157157 240246 182188 108126
barbados ind#743 , 203243 111121 216224 231231 126126 190192 157159 234249 188198 118126
barbados ind#744 , 203225 121135 206232 231231 126130 192198 157159 240249 182222 122156
barbados ind#745 , 203211 121123 224236 227231 126132 192192 157163 238238 182204 138148
barbados ind#746 , 227233 119123 216234 227233 126130 190190 157159 236245 186188 124142
barbados ind#747 , 203223 133141 236240 231231 126126 192198 157157 234236 194204 118126
barbados ind#748 , 203217 141145 222226 227235 126126 190192 145157 238240 182196 108108
barbados ind#749 , 205221 125147 210238 223231 126132 192198 157157 234234 188200 106148
barbados ind#750 , 203235 123123 230234 223231 126132 192198 157157 236256 188214 122136
barbados ind#751 , 203211 121143 212248 223229 126132 190196 157159 234242 214216 118128
barbados ind#752 , 213217 123169 204234 223231 126134 190198 157163 236248 193224 142148
```

Free text

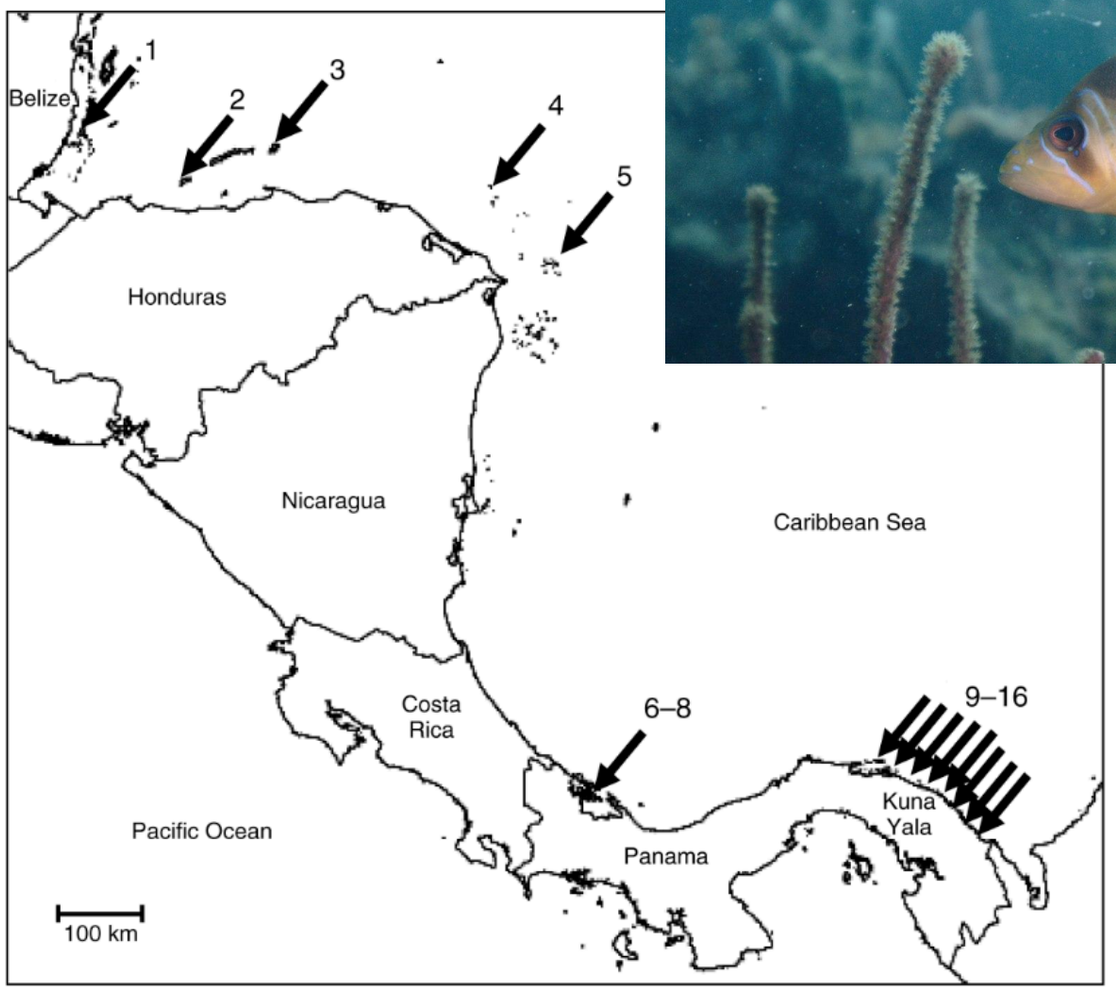
Locus names

Sample / population

Alleles (000 = missing data)

Barred hamlet (*H. puella*) microsatellite dataset

Exercise 3



Ecology, 90(11), 2009, pp. 3087–3098
© 2009 by the Ecological Society of America

Estimating dispersal from genetic isolation by distance in a coral reef fish (*Hypoplectrus puella*)

Oscar Puebla,[1,2,3] Eldredge Bermingham,[1,2] and Frédéric Guichard[2]

- 10 highly variable microsatellite loci
- 854 individuals
- 15 Caribbean locations

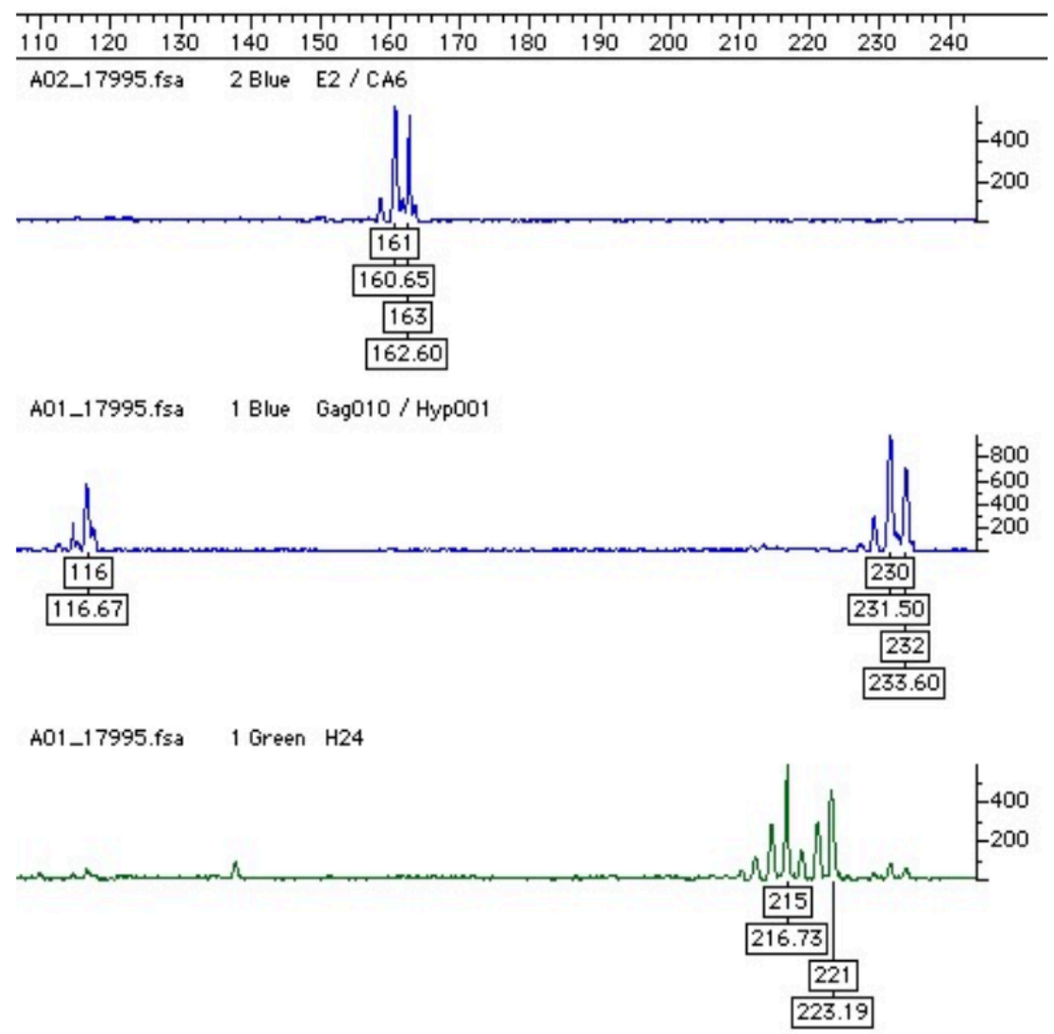
Microsatellites

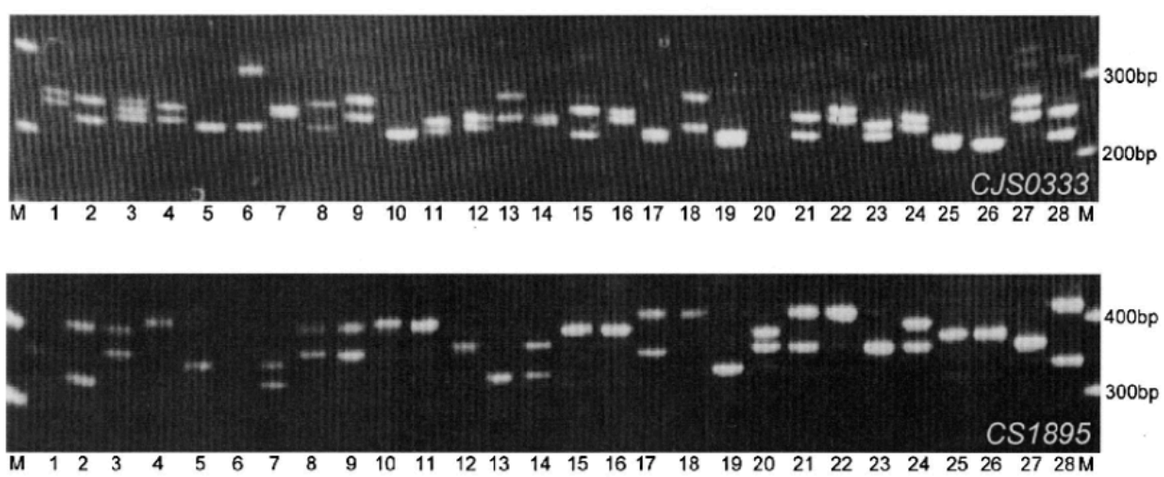
Exercise 3

ACAGTTACAGCAGGAAGTTC […] TATATATATATATATATATATA […] CGCTAATG ACAGCACGCTAAC

$(TA)_n$

- Co-dominant
- Many alleles
- High mutation rates
- High heterozygosities





Useful R functions for population genetics

Summary

Import Genepop file as genind object

```
x <- read.genepop(path_to_file, ncode = n)                    # adegenet package
```

Object name

Name of Genepop file (including path)

No. of characters encoding each allele

Locus statistics (e.g. no. of alleles, Simpson's index, heterozygosity, evenness)

```
locus_table(x)                                                # poppr package
```

Useful R functions for population genetics

Summary

Test for departure from Hardy-Weinberg equilibrium

```
hw.test(x)                                          # quick summary (pegas package)

test_HW(path_to_file, outputFile = new_file)        # detailed (genepop package)
```

Key takeaways

Summary

- **Assumptions matter:** HWE assumes random mating, no selection, and infinite population size etc.; deviations from these assumptions can be biologically informative
- **Locus-specific deviations:** Deviations from HWE expectations often vary by locus, and may indicate locus-specific processes such as natural selection or mutation
- **Genome- and population-wide patterns:** patterns across many loci can reflect broader processes, such as population structure, inbreeding, or demographic history
- **Scientific rigor:** Good data quality is essential to avoid bias from e.g. genotyping error; testing many loci requires correction for multiple comparisons (e.g., Bonferroni)